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5: //Date: 18/1/2024
6:
7: #include<iostream>
8: #include<fstream>
9: #include<string>
10: #include<iomanip>
11: using namespace std;
12:
13: void calconvert(float carbo[], float protein[], float fat[]);
14: void calDV(float carboDV[], float proteinDV[], float fatDV[]);
15: void category(float carboDV[], float proteinDV[], float fatDV[], string level_carbo[], string level_protein[], string level_fat[]);
16: string categoryAVG(float);
17: int main()
18: {
19:     fstream infile("cereal.txt", ios::in );
20:     fstream outfile("cerealoutput.txt", ios::out);
21:     string cerealname[10];
22:     char cerealttype[10];
23:     float carbo[10];
24:     float protein[10];
25:     float fat[10];
26:     float carboDV[10];
27:     float proteinDV[10];
28:     float fatDV[10];
29:     int nutrientchoice,level,num = 1;
30:     float total_carbo = 0;
31:     float total_protein = 0;
32:     float total_fat = 0;
33:     float average_carbo;
34:     float average_protein;
35:     float average_fat;
36:     string level_carbo[10];
37:     string level_protein[10];
38:     string level_fat[10];
39:     char choice;
40:     string level_avgcarb;
41:     string level_avgpro;
42:     string level_avgfat;
43:
44:     if(!infile)
45:     {
46:         cout<<"ERROR: Cannot open file\n";
47:         exit(1);
48:     }
49:     for(int i=0; i<10; i++)
50:     {
51:         infile>>cerealname[i]>>cerealttype[i]>>carbo[i]>>protein[i]>>fat[i];
52:     }
53:     do
54:     {

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55:     cout<<"1.Carbohydrate\n2.Protein\n3.Fat\n";
56:     cout<<"Please enter your choice>> ";
57:     cin>>nutrientchoice;
58:     cout<<"\n1.Low\n2.Moderate\n3.High\n";
59:     cout<<"Please enter your choice>> ";
60:     cin>>level;
61:     cout<<endl;
62:
63:     calconvert(carbo, protein, fat);
64:
65:     for(int i=0; i<10;i++)
66:     {
67:         carboDV[i]=carbo[i];
68:         proteinDV[i]=protein[i];
69:         fatDV[i]=fat[i];
70:     }
71:
72:     calDV( carboDV, proteinDV, fatDV);
73:     category(carboDV, proteinDV, fatDV, level_carbo, level_protein, level_fat);
74:
75:     //Determine the average %DV
76:     for(int i=0; i<10;i++)
77:     {
78:         total_carbo = total_carbo + carboDV[i];
79:         total_protein = total_protein + proteinDV[i];
80:         total_fat = total_fat + fatDV[i];
81:     }
82:
83:     //Arithmetic operation
84:     average_carbo = total_carbo / 10;
85:     average_protein = total_protein / 10;
86:     average_fat = total_fat / 10;
87:
88:     if(nutrientchoice==1)
89:     {
90:         if(level == 1)
91:         {
92:             cout<<"List of cereal with low amount of carbohydrate:\n";
93:             for(int i=0 ; i<10; i++)
94:             {
95:                 if(level_carbo[i] == "Low")
96:                 {
97:                     cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
98:                     num++;
99:                 }
100:            }
101:        }
102:        if(level == 2)
103:        {
104:            cout<<"List of cereal with moderate amount of carbohydrate:\n";
105:            for(int i=0 ; i<10; i++)
106:            {
107:                if(level_carbo[i] == "Moderate")
108:                {

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109:         cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
110:         num++;
111:     }
112: }
113: }
114: if(level == 3)
115: {
116:     cout<<"List of cereal with high amount of carbohydrate:\n";
117:     for(int i=0 ; i<10; i++)
118:     {
119:         if(level_carbo[i] == "High")
120:         {
121:             cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
122:             num++;
123:         }
124:     }
125: }
126: }
127:
128: if(nutrientchoice==2)
129: {
130:     if(level == 1)
131:     {
132:         cout<<"List of cereal with low amount of protein:\n";
133:         for(int i=0 ; i<10; i++)
134:         {
135:             if(level_carbo[i] == "Low")
136:             {
137:                 cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
138:                 num++;
139:             }
140:         }
141:     }
142:     if(level == 2)
143:     {
144:         cout<<"List of cereal with moderate amount of protein:\n";
145:         for(int i=0 ; i<10; i++)
146:         {
147:             if(level_carbo[i] == "Moderate")
148:             {
149:                 cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
150:                 num++;
151:             }
152:         }
153:     }
154:     if(level == 3)
155:     {
156:         cout<<"List of cereal with high amount of protein:\n";
157:         for(int i=0 ; i<10; i++)
158:         {
159:             if(level_carbo[i] == "High")
160:             {
161:                 cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
162:                 num++;

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163:     }
164: }
165: }
166: }
167: if(nutrientchoice==3)
168: {
169:     if(level == 1)
170:     {
171:         cout<<"List of cereal with low amount of fat:\n";
172:         for(int i=0 ; i<10; i++)
173:         {
174:             if(level_carbo[i] == "Low")
175:             {
176:                 cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
177:                 num++;
178:             }
179:         }
180:     }
181:     if(level == 2)
182:     {
183:         cout<<"List of cereal with moderate amount of fat:\n";
184:         for(int i=0 ; i<10; i++)
185:         {
186:             if(level_carbo[i] == "Moderate")
187:             {
188:                 cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
189:                 num++;
190:             }
191:         }
192:     }
193:     if(level == 3)
194:     {
195:         cout<<"List of cereal with high amount of fat:\n";
196:         for(int i=0 ; i<10; i++)
197:         {
198:             if(level_carbo[i] == "High")
199:             {
200:                 cout<<num<<" . Cereal "<<cerealtype[i]<<endl;
201:                 num++;
202:             }
203:         }
204:     }
205:     level_avgcarb = categoryAVG(average_carbo);
206:     level_avgpro = categoryAVG(average_protein);
207:     level_avgfat = categoryAVG(average_fat);
208: }
209: cout<<"Choose again? Y/N >> ";
210: cin>>choice;
211:
212: }while((choice == 'Y')||(choice == 'y'));
213:
214: outfile<<"Type\t\tCarbohydrate\t\t Protein\t\t Fat\n";
215: outfile<<"=====\t=====\t\t =====\t\t ===\n";
216: for(int i=0;i<10;i++)

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217: {
218:     outfile<<"Cereal "<<cerealtype[i]<<"\t";
219:     if (level_carbo[i] == "Moderate")
220:         outfile<<fixed<<setprecision(2)<<setw(5)<<carboDV[i]<<"%("<<level_carbo[i]<<left<<setw(10)<<"");
221:     else if (level_carbo[i] == "Low")
222:         outfile<<fixed<<setprecision(2)<<setw(5)<<carboDV[i]<<"%("<<level_carbo[i]<<left<<setw(15)<<"");
223:     else
224:         outfile<<fixed<<setprecision(2)<<setw(5)<<carboDV[i]<<"%("<<level_carbo[i]<<left<<setw(14)<<"");
225:
226:     if(level_protein[i] == "Moderate")
227:     {
228:         outfile<<setw(5)<<proteinDV[i]<<"%("<<level_protein[i]<<left<<setw(10)<<"");
229:     }
230:     else if(level_protein[i] == "Low")
231:         outfile<<setw(5)<<proteinDV[i]<<"%("<<level_protein[i]<<left<<setw(15)<<"");
232:     else
233:         outfile<<setw(5)<<proteinDV[i]<<"%("<<level_protein[i]<<left<<setw(14)<<"");
234:
235:     outfile<<fatDV[i]<<"%("<<level_fat[i]<<"")\n";
236: }
237: outfile<<"The ten cereal types produce an average %DV of:\n";
238: outfile<<"Carbohydrate: "<<average_carbo<<"%("<<level_avgcarb<<"")\n";
239: outfile<<"Protein: "<<average_protein<<"%("<<level_avgpro<<"")\n";
240: outfile<<"Fat: "<<average_fat<<"%("<<level_avgfat<<"")\n";
241:
242: infile.close();
243: outfile.close();
244:
245: return 0;
246: }
247: void calconvert(float carbo[], float protein[], float fat[])
248: {
249:     for(int i=0; i<10; i++)
250:     {
251:         carbo[i] = carbo[i]*28.35;
252:         protein[i] = protein[i]*28.35;
253:         fat[i] = fat[i]*28.35;
254:     }
255: }
256: void calDV(float carboDV[], float proteinDV[], float fatDV[])
257: {
258:     for(int i=0; i<10;i++)
259:     {
260:         carboDV[i] = carboDV[i]/300*100;
261:         proteinDV[i] = proteinDV[i]/50*100;
262:         fatDV[i] = fatDV[i]/65*100;
263:     }
264: }
265:
266: void category(float carboDV[], float proteinDV[], float fatDV[], string level_carbo[], string level_protein[], string level_fat[])
267: {
268:     //Loop for carboDV[]
269:     for(int i=0; i<10;i++)
270:     {

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271:         if (carboDV[i] <= 5)
272:             level_carbo[i]= "Low";
273:
274:         else if ((carboDV[i] > 5) && (carboDV[i] < 20))
275:             level_carbo[i]= "Moderate";
276:
277:         else
278:             level_carbo[i]= "High";
279:     }
280:
281:     //Loop for proteinDV[]
282:     for(int i=0; i<10;i++)
283:     {
284:         if (proteinDV[i] <= 5)
285:             level_carbo[i]= "Low";
286:
287:         else if ((proteinDV[i] > 5) && (proteinDV[i] < 20))
288:             level_protein[i]= "Moderate";
289:
290:         else
291:             level_protein[i]= "High";
292:     }
293:
294:     //Loop for fatDV[]
295:     for(int i=0; i<10;i++)
296:     {
297:         if (fatDV[i] <= 5)
298:             level_fat[i]= "Low";
299:
300:         else if ((fatDV[i] > 5) && (fatDV[i] < 20))
301:             level_fat[i]= "Moderate";
302:
303:         else
304:             level_fat[i]= "High";
305:     }
306: }
307: string categoryAVG(float avg)
308: {
309:     string level;
310:
311:     if (avg <= 5)
312:         level= "Low";
313:
314:     else if ((avg > 5) && (avg < 20))
315:         level = "Moderate";
316:
317:     else
318:         level= "High";
319:
320:     return level;
321: }

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